

Practice problems #21

1. Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at $(1, 4, 0)$, $(-2, -5, 2)$, $(-1, 2, -1)$.
2. Let R be the triangle with vertices at (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) . Show that area of the triangle R equals $\frac{1}{2} \left| \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \right|$.
3. Let $T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear transformation that maps e_1, e_2, e_3 to ae_1, be_2, ce_3 , respectively, where a, b and c are some positive real numbers. Let S be the unit ball in $\mathbf{R}^3$ bounded by the unit sphere $x^2 + y^2 + z^2 = 1$. Use the fact that the volume of the unit ball is $4\pi/3$ to find the volume of $T(S)$.

Answers or Hints are on page 2.

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#2: The area of R is $\frac{1}{2} \left| \det \begin{bmatrix} x_1 - x_3 & x_2 - x_3 \\ y_1 - y_3 & y_2 - y_3 \end{bmatrix} \right|$, and the matrix $\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$ is row

equivalent to the matrix $\begin{bmatrix} x_1 - x_3 & y_1 - y_3 & 0 \\ x_2 - x_3 & y_2 - y_3 & 0 \\ x_3 & y_3 & 1 \end{bmatrix}$.

#3: $abc4\pi/3$